ENGINEERING ECONOMICS

LECTURE - 01

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Engineering

Engineering is defined as finding solution to the day today problems by application of Mathematics, Natural and other Sciences using knowledge and experience to develop the products and services E*conomically*

Natural Sciences

The sciences involved in the study of the physical world and its phenomena (developments)

Social Sciences

The branch of science that studies society and the relationships of individual within a society.

<u>Economics</u>

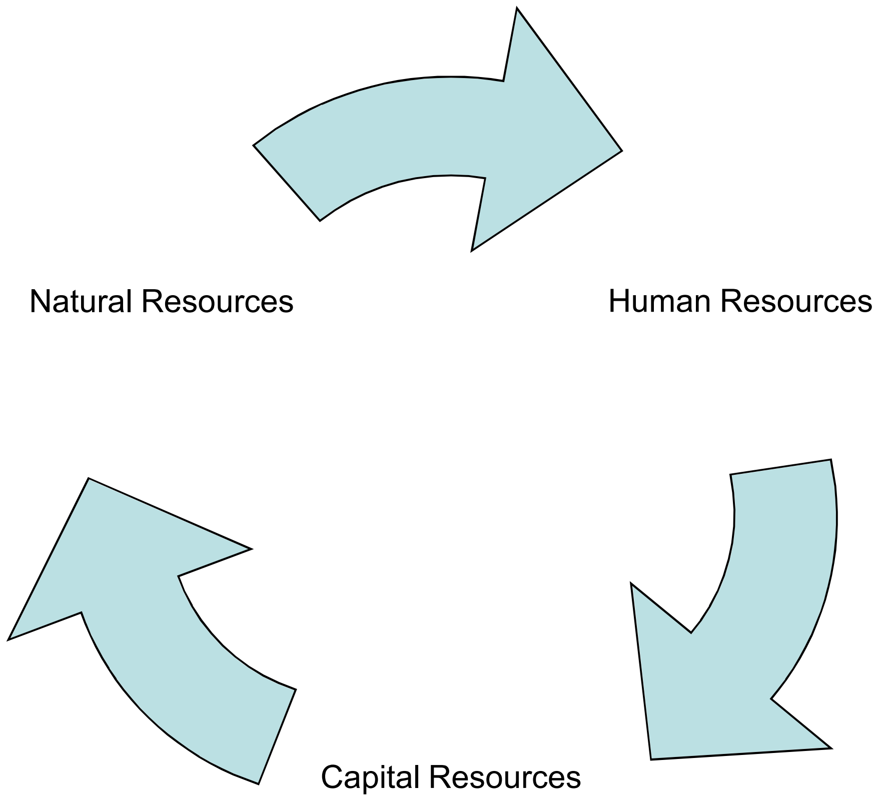
- The branch of social science that deals with the production, distribution and consumption of goods and services efficiently and effewctively.

- The study of how limited resources are used to satisfy unlimited human wants.

- The study of how individuals and societies <u>*choose*</u> to ration limited resources that nature and previous generations have provided.

Resources

Three types of resources work together in our economy



Note: Read word file (economic resource) in folder (other)

<u>Needs vs. Wants</u>

- Needs – what people must have to live.
  - Food
  - Clothing
  - Shelter

- Wants – the things we would like to have, but can live without.

Economy

1. The system of production, distribution and consumption
2. The efficient use of resources

System A group of independent but interrelated elements comprising a unified whole.

Viability Capable of being done in a practical and useful way

Alternative An alternative is a stand alone solution for a given thing

Engineering Economics

Engineering Economics is a subset of economics for application to engineering.

Engineers seek solutions to problems, and the economic viability of each potential solution is normally considered along with the technical aspects.

Engineering Economy

It is the collection of techniques that simplify the comparisons of alternatives on economic basis.

Why Study Engineering Economics?

- There is a traditional perception stating that engineers deal with technical part of projects and financial matters are left to economists and financial analysts.

- Nowadays, the situation is different. Engineers are not limited to their conventional (core) activities. Their role is expanding. They are actively involved in strategic and operational decision making processes of their companies.

- On top of that many engineers are expected to establish their own firms. In this scenario, they need to acquire broad knowledge to run their companies effectively and efficiently.
- Therefore, learning the concepts, techniques and methods of engineering economics can help them to analyze initiated projects from the economic point of view.
- In reality, any engineering project must be not only physically realizable but also economically affordable.

For example, a child tricycle could be built with an aluminum frame or a composite frame.

Some may argue that the composite frame will be stronger and lighter, it is a better choice.

However, there is not much of a market for thousand dollar tricycle.

This scenario reinforces the idea that the economic factors of a design weigh heavily in the design process, and that engineering economics is an integral part of that process, regardless of the engineering discipline.

Role in Capital Investment Decisions

- Engineers play a major role in capital investment decisions based on their analysis, synthesis and design efforts.

- The factors considered in making the decision are a combination of economic and non economic factors.

- Additional factors may be intangible, such as convenience, goodwill, friendship and others.

- Fundamentally, engineering economics involves formulating, estimating, and evaluating the economic outcomes when alternatives to accomplish a defined purpose are available.

Examples of Capital Investment Decisions for Engineering Activities:

- Should a new bonding technique be incorporated into the manufacture of automobile brake pads?
- If a computer-vision system replaces the human inspector in performing quality tests on an automobile welding line, will operating costs decrease over a time horizon of 5 years?
- Is it an economically wise decision to upgrade the composite material production center of an airplane factory in order to reduce costs by 20%?

- Should a highway bypass be constructed around a city of 25,000 people?
- Should the current roadway through the city be expanded?
- Will we make the required rate of return if we install the newly offered technology onto our medical laser manufacturing line?

BI Enviromental Nature of Engineering

Physical Environment :
Engineers produce products and services depending on physical laws (e.g. Newton's laws).

Economic Environment :
Much less of a quantitative nature is known about economic environments -- this is due to economics being involved with the actions of people, and the structure of organizations.

Satisfaction of the physical and economic environments is linked through production and construction processes.

Engineers need to manipulate systems to achieve a balance in attributes in both the physical and economic environments, and within the bounds of limited resources.

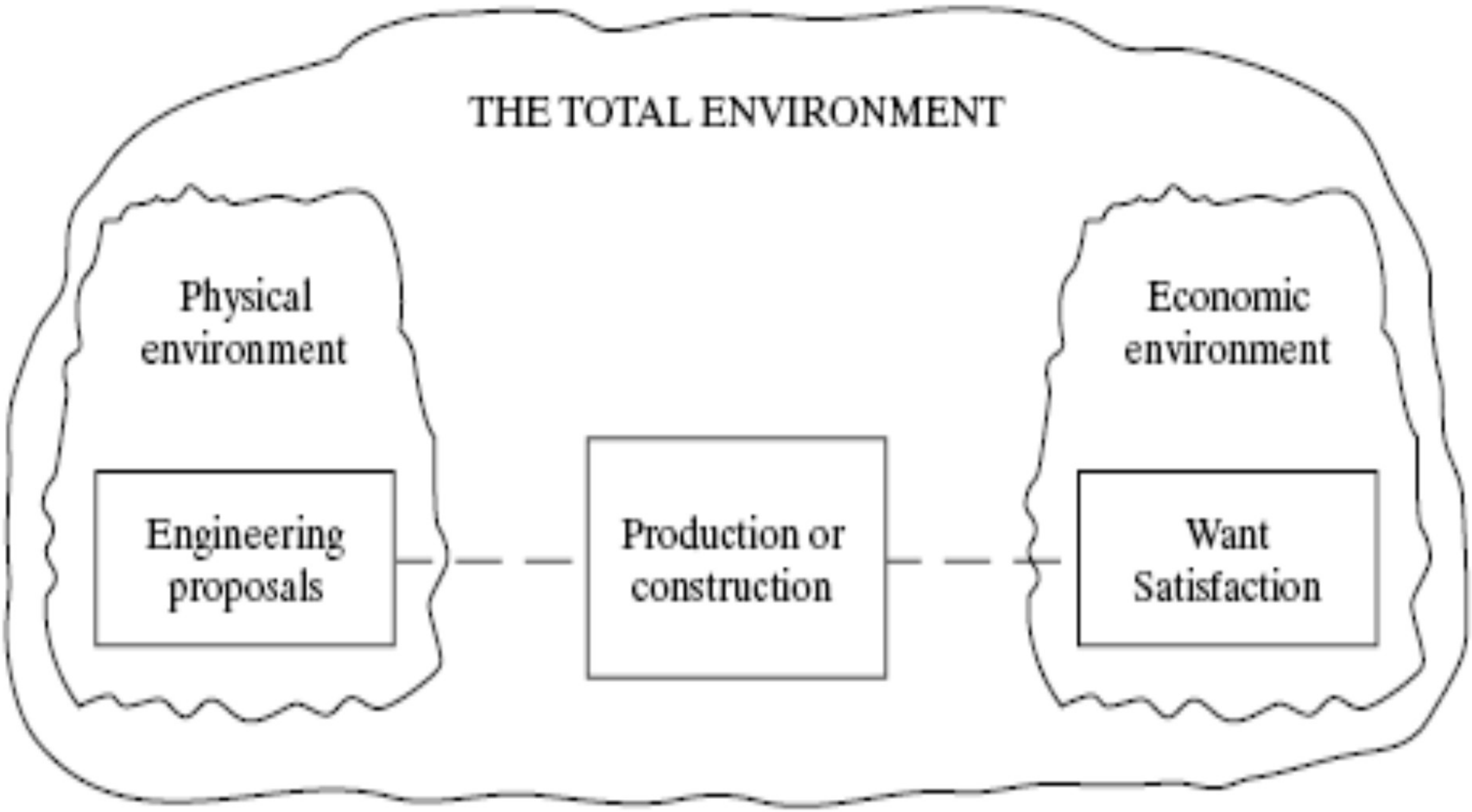


Figure 1.1 : Physical and Economic Components of an Engineering System

Figure 1.1 shows how engineering is composed of physical and economic components.

Discussion